

Wheel Selection Menu

How Should Our Robot Touch the Field?

Graduate Expectation Tags

- **[Purple: Curiosity & Creativity]** We explore different wheel types and imagine how they affect robot movement.
- **[Red: Critical Thinking & Problem Solving]** We compare traction, turning, speed, control, and reliability.
- **[Green: Personal Development]** We make responsible choices based on available materials and team skill.
- **[Blue: Effective Communication]** We explain wheel choices using sketches, test data, and observations.

Decision Being Made

Your team is choosing the wheels for the chassis. Wheels affect how the robot drives, turns, accelerates, pushes, stops, and handles contact with the field or game objects.

Teams may have access to wheels from different systems, including Tetrix, REV, goBILDA, and others. Wheel sizes, tread patterns, rubber density, hubs, axles, and connectors may all be different.

Background Sources

Use ideas from:

- **Chassis Shape Menu**
- **Chassis Material Menu**
- **Chassis Physics and Space Planning**
- robot driving observations,
- available wheel bins,
- mentor demonstrations,
- and your team's chassis sketch.

Why This Decision Matters

Wheels are not just "the things that roll." They determine how the robot interacts with the floor.

Wheel choice affects:

- traction,
- turning,
- speed,
- stability,
- pushing ability,
- driver control,
- autonomous consistency,

- clearance,
- and how easily the wheels attach to motors, hubs, shafts, and the chassis.

A wheel that looks good may not fit the shaft. A wheel that grips well may make turning harder. A wheel that turns easily may not push well. Your team needs to choose based on the job the robot needs to do.

How to Use This Menu

- Review the wheel options.
- Check which wheels are actually available.
- Check whether the wheels fit your hubs, shafts, axles, and chassis.
- Decide which wheels are powered and which wheels support the robot.
- Sketch the wheel layout.
- Record your choice, reasoning, trade-off, and evidence plan in the portfolio.

Wheel Design Factors

Wheel Size

Larger wheels can move the robot farther with each rotation, which can make the robot faster. They can also increase clearance off the floor. Smaller wheels may be easier to fit and may give more control.

Larger Wheels	Smaller Wheels
More distance per rotation	More compact
More floor clearance	Often easier to fit
May increase speed	May feel easier to control
May need more torque	May reduce clearance

Tread and Rubber

Soft rubber and grippy tread can help the robot push and accelerate. Harder wheels or smoother wheels may slip more but can sometimes turn more easily.

More Grip	Less Grip
Better pushing and acceleration	Easier turning in some layouts
More control when driving straight	Less strain during turns
May resist sideways movement	May slip under load
Can make turning harder	May reduce pushing power

Fit and Connection

Before choosing a wheel, check:

- Does the wheel fit the hub?

- Does the hub fit the shaft or axle?
- Is the shaft round or hex?
- Is the set screw tight?
- Is the wheel supported?
- Does the wheel rub the frame?
- Do we have spacers to stop side-to-side sliding?
- Does the wheel match the motor placement and chassis shape?

Menu Options

Option A: Standard Traction Wheels

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: Traction wheels are regular wheels designed to grip the floor.

What it helps: They provide pushing power, acceleration, and strong forward/backward movement.

Best for: Drive wheels powered by motors.

Ease of use: Easy to medium

Likelihood of success: High if they fit the hub and shaft

Time to implement: Fast to moderate

Material / tooling needs: Wheels, hubs, shafts/axles, set screws, spacers, bearings or supports, and hex tools.

Possible trade-offs: Too much grip can make turning harder, especially if all four wheels are traction wheels.

Evidence we might watch: Does the robot drive straight? Does it turn smoothly? Do the wheels slip or scrub during turns?

Option B: Omni Wheels

Graduate Expectation Connection: [Purple: Curiosity & Creativity] and [Red: Critical Thinking & Problem Solving]

What it means: Omni wheels have small rollers around the edge. They roll forward like normal wheels, but can slide sideways more easily.

What it helps: Omni wheels reduce sideways resistance during turns. A common beginner layout uses two powered traction wheels and two omni support wheels.

Best for: Helping a two-motor robot turn more smoothly.

Ease of use: Medium

Likelihood of success: High if installed correctly

Time to implement: Moderate

Material / tooling needs: Omni wheels, compatible hubs, shafts/axles, spacers, and supports.

Possible trade-offs: Omni wheels usually provide less sideways resistance, so the robot may be easier to push sideways. They may provide less traction than regular wheels.

Evidence we might watch: Does the robot turn more easily? Does it drift sideways? Does it still drive straight enough for testing?

Option C: Four Traction Wheels

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: All four wheels are regular traction wheels.

What it helps: This can give the robot strong grip and pushing power.

Best for: Robots that need to push or hold position and do not need to turn very tightly.

Ease of use: Easy

Likelihood of success: Medium

Time to implement: Fast

Material / tooling needs: Four matching traction wheels, hubs, axles/shafts, spacers, and supports.

Possible trade-offs: The robot may fight itself while turning because all four wheels resist sideways sliding. This is called wheel scrub. It can make turning jerky, slow, or hard on motors.

Evidence we might watch: Does the robot turn smoothly? Do motors strain during turns? Does the robot jerk or skid?

Option D: Two Traction Wheels + Two Omni Wheels

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: Two powered traction wheels move the robot. Two omni wheels support the robot and help it turn.

What it helps: This is a strong beginner layout because it balances powered movement with easier turning.

Best for: Two-motor course robots that need simple, reliable driving.

Ease of use: Medium

Likelihood of success: High

Time to implement: Moderate

Material / tooling needs: Two traction wheels, two omni wheels, compatible hubs/shafts/spacers, and supports.

Possible trade-offs: Less pushing power than four traction wheels. The robot may slide more easily from the omni-wheel side.

Evidence we might watch: Does the robot turn predictably? Does it drive straight? Can the driver control it easily?

Option E: Mecanum Wheels

Graduate Expectation Connection: [Purple: Curiosity & Creativity] and [Red: Critical Thinking & Problem Solving]

What it means: Mecanum wheels have angled rollers that allow a robot to drive forward/backward, turn, and move sideways when used in a four-motor drivetrain.

What it helps: Mecanum drive can make it easier to line up with goals, move around field objects, and strafe sideways.

Best for: Advanced teams with enough motors, wheels, time, and programming support.

Ease of use: Hard

Likelihood of success: Risky for a short course unless the base already exists

Time to implement: Slow

Material / tooling needs: Four mecanum wheels, four drive motors, correct wheel orientation, motor controllers, more complex code, tuning, and driver practice.

Possible trade-offs: More expensive, more complex, more programming, more wiring, and more debugging. It may take time away from shooter, hopper, trigger, portfolio, and testing.

Evidence we might watch: Can the robot strafe reliably? Do all wheels spin in the correct direction? Can the driver control it under pressure?

Option F: Reused Wheels From an Existing Chassis

Graduate Expectation Connection: [Green: Personal Development] and [Orange: Cultural Understanding & Civic Engagement]

What it means: The team uses wheels already mounted on a known chassis or salvages wheels from an existing robot base.

What it helps: This can save time and reduce fit problems because the wheels, hubs, shafts, and supports may already work together.

Best for: Teams that want to prioritize robot reliability, programming, mechanisms, or testing time.

Ease of use: Easy to medium

Likelihood of success: High if the wheels are in good condition

Time to implement: Fast

Material / tooling needs: Existing chassis or wheel assemblies, inspection checklist, hex tools, replacement screws, spacers, or hubs if needed.

Possible trade-offs: The wheel layout may limit chassis shape, motor placement, ground clearance, or future mechanisms.

Evidence we might watch: Do the wheels spin freely? Are set screws tight? Does the robot drive straight and turn reliably?

Wheel Layout Questions

Before finalizing the wheel choice, answer:

- Which wheels are powered by motors?
- Which wheels are support wheels?
- Where are the omni wheels, if used?
- Do the wheels fit the hubs and shafts?
- Are the set screws tight?
- Is the wheel supported on the axle?
- Does the wheel rub the frame?

- Is there enough clearance off the floor?
- Will the robot turn smoothly?
- Will the driver be able to control the robot?

Wheel Assembly Checklist

Check	Yes / No
Wheel fits the hub	
Hub fits the shaft or axle	
Set screw is tight	
Wheel spins freely	
Wheel does not rub the frame	
Spacers stop side-to-side sliding	
Axle or shaft is supported	
Left and right drive wheels are aligned	
Omni wheels roll in the correct direction	
Robot has enough floor clearance	

Comparison Table

Option	Best Feature	Main Risk	Good For
Standard traction wheels	Grip and pushing	Turning may be harder	Powered drive wheels
Omni wheels	Easier turning	Less sideways resistance	Support wheels on simple drivetrains
Four traction wheels	Strong grip	Wheel scrub during turns	Pushing or holding position
Two traction + two omni	Balanced beginner drive	Less pushing than four traction	Reliable two-motor robot
Mecanum wheels	Sideways movement	Complex, expensive, needs four motors	Advanced competition-style driving
Reused wheel assemblies	Saves time	Limits design choices	Reliable fast start

Wheel Decision Table

Record your decision in your engineering portfolio.

Decision Area	Our Choice	Why We Chose It	What We Gave Up	Evidence We Will Watch
Drive wheels				

Decision Area	Our Choice	Why We Chose It	What We Gave Up	Evidence We Will Watch
Support wheels				
Wheel size				
Wheel material/tread				
Wheel layout				

Must / Should / Could

Level	Expectation
Must Do	Choose wheels that fit the chassis and identify which wheels are powered.
Should Do	Explain how the wheel choice affects turning, traction, control, and reliability.
Could Do	Compare two wheel layouts using a drive test, turning test, or driver feedback.

SLD Connection

Main SLD Prompt

Should our robot prioritize traction, easy turning, speed, or driver control when choosing wheels?

Supporting Questions

- What wheel choice gives us the best chance of driving reliably?
- When is more grip helpful?
- When can too much grip make turning harder?
- Should we use omni wheels to make turning easier?
- Is mecanum worth the extra complexity in a short course?
- Should we reuse a known wheel setup to save time?

Portfolio Deliverable

Add the following to your engineering portfolio:

- selected wheel types,
- powered wheel locations,
- support wheel locations,
- wheel layout sketch,
- wheel assembly checklist,

- explanation of the main trade-off,
- and evidence the team will watch during testing.

Reflection

Our wheel choice is:

We chose this because:

The main trade-off we accepted is:

The evidence we will watch during testing is:
